

A Simple KFBIM Shape Optimization Problem

KFBIM notes

May 9, 2026

Goal

The application `shape_opt_transmission_2d` solves a small inverse shape problem for a two-phase screened diffusion equation. The unknown is the boundary $\Gamma(a)$ of an inclusion $D(a) \subset \Omega$, where $\Omega = (-1.2, 1.2)^2$. Synthetic target data are generated from a known hidden shape, and the optimization attempts to recover that shape from exterior field measurements.

Forward Transmission Problem

For each outer boundary excitation g_ℓ , solve

$$-\nabla \cdot (\beta \nabla u_\ell) + \kappa^2 u_\ell = 0 \quad \text{in } \Omega \setminus \Gamma(a), \quad (1)$$

$$u_\ell = g_\ell \quad \text{on } \partial\Omega, \quad (2)$$

$$[u_\ell] = 0 \quad \text{on } \Gamma(a), \quad (3)$$

$$[\beta \partial_n u_\ell] = 0 \quad \text{on } \Gamma(a). \quad (4)$$

The coefficient is piecewise constant:

$$\beta = \begin{cases} \beta_{\text{int}}, & x \in D(a), \\ \beta_{\text{ext}}, & x \in \Omega \setminus \overline{D(a)}. \end{cases}$$

The implemented demo uses

$$\beta_{\text{int}} = 4, \quad \beta_{\text{ext}} = 1, \quad \frac{\kappa_{\text{int}}^2}{\beta_{\text{int}}} = \frac{\kappa_{\text{ext}}^2}{\beta_{\text{ext}}} = \lambda^2 = 1.1.$$

Thus the KFBIM common-ratio transmission API can reduce both phases to the same screened operator $-\Delta + \lambda^2$ after division by β .

The boundary excitations are

$$g_0 = 1, \quad g_1 = x, \quad g_2 = y, \quad g_3 = x^2 - y^2.$$

For a candidate shape, KFBIM returns the grid solution values $u_\ell(x_m; a)$ at fixed observation nodes x_m in an exterior annulus.

Shape Parameterization

The inclusion boundary is represented as a fixed-center radial Fourier curve:

$$\Gamma(a) : \quad X(\theta; a) = c + R(\theta; a)(\cos \theta, \sin \theta), \quad c = (0.07, -0.04),$$

with

$$R(\theta; a) = R_0 \exp \left(\sum_{k=1}^K a_k \cos(k\theta) + b_k \sin(k\theta) \right), \quad R_0 = 0.5, \quad K = 3.$$

The exponential form keeps the radius positive. The hidden target used by the demo is

$$(a_1, b_1, a_2, b_2, a_3, b_3) = (0.04, 0, 0, -0.08, 0.16, 0).$$

Optimization Objective

Let $d_{\ell m} = u_{\ell}(x_m; a_{\star})$ denote synthetic measurements from the target shape $a_{\star}$. The demo minimizes

$$J(a) = \frac{1}{2M} \sum_{\ell, m} (u_{\ell}(x_m; a) - d_{\ell m})^2 + \alpha \sum_{k=1}^K k^4 (a_k^2 + b_k^2) + \rho (|D(a)| - |D(a_{\star})|)^2.$$

The first term measures mismatch to the target data. The second term is a smoothness regularizer that penalizes high Fourier modes. The third term discourages area drift.

Adjoint Shape Gradient

The optimized application can replace finite-difference gradients by one continuous adjoint solve per boundary excitation. For each forward solution u_{ℓ} , the adjoint p_{ℓ} solves

$$-\nabla \cdot (\beta \nabla p_{\ell}) + \kappa^2 p_{\ell} = \frac{1}{M_{\text{tot}}} \sum_m (u_{\ell}(x_m; a) - d_{\ell m}) \delta_{x_m},$$

with homogeneous Dirichlet data on $\partial\Omega$ and the same transmission conditions

$$[p_{\ell}] = 0, \quad [\beta \partial_n p_{\ell}] = 0 \quad \text{on } \Gamma(a).$$

On the Cartesian grid, each point source is represented by a nodal delta with weight proportional to $1/h^2$. Since the KFBIM transmission API takes the reduced right-hand side $q = f/\beta$, the exterior observation source is entered as

$$q(x_m) = \frac{u_{\ell}(x_m; a) - d_{\ell m}}{M_{\text{tot}} \beta_{\text{ext}} h^2}.$$

Let n be the normal pointing from the inclusion to the exterior and let τ be a unit tangent. Because both the state and adjoint are continuous across the interface, their tangential derivatives are single-valued. The implemented continuous shape-gradient density is

$$g_{\Gamma} = \sum_{\ell} \left\{ \beta_{\text{int}} \partial_n u_{\ell, \text{int}} \partial_n p_{\ell, \text{int}} - \beta_{\text{ext}} \partial_n u_{\ell, \text{ext}} \partial_n p_{\ell, \text{ext}} - (\beta_{\text{int}} - \beta_{\text{ext}}) [\partial_{\tau} u_{\ell} \partial_{\tau} p_{\ell} + \lambda^2 u_{\ell} p_{\ell}] \right\}.$$

For a Fourier coefficient, this density is projected onto the normal velocity generated by that coefficient. For example,

$$\frac{\partial J_{\text{data}}}{\partial a_k} = \int_{\Gamma} g_{\Gamma} R(\theta) \cos(k\theta) (e_r \cdot n) ds,$$

and the formula for b_k replaces $\cos(k\theta)$ by $\sin(k\theta)$. The regularization and area-penalty derivatives are then added explicitly.

Numerical Method

For each trial vector a , the code:

1. builds a P2 quadratic interface with `CurveResampler2D::discretize`;
2. solves the four common-ratio transmission problems using `LaplaceTransmission2D`;
3. samples the resulting fields at the observation nodes;
4. evaluates $J(a)$.

The default implementation now uses the adjoint gradient with a backtracking gradient-descent step. The finite-difference gradient remains available for verification. A gradient check compares the adjoint direction against central finite differences at the current shape.

Important Practical Point

If the regularization weight α is too large, the exact target shape may not minimize the chosen objective. In the default target, the $k = 3$ mode is important, but the regularizer weights it by $3^4 = 81$. For clean synthetic recovery, use a very small α or set $\alpha = 0$. For noisy data, keep a small positive α to suppress nonphysical high-frequency shape changes.

Research-Level Direction

The demo is intentionally small, but it points to a larger class of inverse interface problems that is well matched to KFBIM: recover a moving embedded biophysical or materials boundary from fields measured away from the boundary. One concrete story is an active cell or organelle whose membrane separates two diffusive media with different transport coefficients. Boundary actuation or external illumination produces several steady screened-diffusion fields, while microscopy or electrodes measure only the exterior response. The inverse problem is to infer the hidden membrane shape, not merely the scalar coefficient values.

This setting is difficult for a standard body-fitted finite-element method because every shape update changes the mesh and the interface geometry. It is also awkward for a classical boundary integral equation when the bulk operator, forcing, boundary conditions, or future variable coefficients destroy the availability of a simple analytic Green's function. KFBIM keeps the interface geometric unknown explicit while using Cartesian bulk solvers, so shape iterations can move the boundary without remeshing the full domain.

KFBIM-Specific Metric

The current app uses a Euclidean gradient in the six Fourier coefficients. A more KFBIM-specific iteration would replace this by a geometry-aware preconditioned gradient. Let $V_i(\theta)$ be the normal velocity generated by parameter a_i or b_i . Instead of stepping with

$$\delta a_i = -\eta \frac{\partial J}{\partial a_i},$$

define a Gram matrix

$$G_{ij} = \int_{\Gamma} V_i V_j ds + \ell^2 \int_{\Gamma} \partial_s V_i \partial_s V_j ds + \gamma \langle \mathcal{K} V_i, \mathcal{K} V_j \rangle_{\Omega_h}.$$

Here $\mathcal{KV}$ denotes the KFBIM linearized field response produced by a normal boundary displacement V . In practice, this response can be estimated with the same spread–bulk–solve–restrict machinery that already applies the PDE operator. The update is then

$$G \delta a = -\nabla_a J.$$

The first two terms give the usual L^2 or Sobolev geometry metric on the boundary. The last term is problem-dependent: two boundary motions are close if KFBIM predicts that they produce similar observable PDE fields. This can reduce wasted steps in nearly invisible shape directions, balance Fourier modes according to the actual measurement map rather than an arbitrary coefficient scale, and make line search less sensitive to the grid resolution or the chosen parameterization.

Why This Can Be Justified

The metric above is a Gauss–Newton or natural-gradient approximation for the reduced objective. If $\mathcal{F}(a)$ maps shape parameters to all observed data, then the data term is

$$J_{\text{data}}(a) = \frac{1}{2} \|\mathcal{F}(a) - d\|^2.$$

Near a local minimizer,

$$\mathcal{F}(a + \delta a) \approx \mathcal{F}(a) + \mathcal{F}'(a)\delta a,$$

so the local quadratic model has Hessian

$$\mathcal{F}'(a)^* \mathcal{F}'(a)$$

plus regularization terms, up to residual-weighted second derivatives. The KFBIM response term in G is a discretized version of this pullback metric. Under the usual differentiability assumptions for smooth elliptic transmission shape functionals, this produces a descent direction whenever G is symmetric positive definite and the adjoint gradient is consistent with the chosen discretization. It is not a magic convexification, but it is a principled preconditioner for the reduced-space optimization.